Candidates select and study **three** topics. At least **one** topic must be chosen fr
Section A: *Environmental Issues* and at least **one** topic must be chosen from
Section B: *Economic Issues*.

## Section A: Environmental Issues

At least **one** topic must be chosen from this section.

**Option A1: Earth hazards**

*Candidates need to have a knowledge and understanding of the meaning of 'hazards', their degree of predictability and that they make both short- and long-term impacts on an environment and community. Case studies should be selected from different areas and at scales appropriate to the hazards being examined.*

| Earth hazards | | |
|---|---|---|
| *Questions for Investigation* | *Key Ideas* | *Content* |
| What are the hazards associated with mass movement and slope failure? | Mass movement is more likely to occur when both physical and human factors disturb the equilibrium of a slope. Mass movement has a range of environmental and social impacts on the areas affected, which create a range of human responses to the hazard. | The study of the processes and conditions that lead to mass movements:<br>• physical conditions (including slope angle, weathering, vegetation, climate and weather, drainage and rock types) and human activities (including deforestation, adding weight, undercutting slopes, quarrying) leading to the various types of mass movement;<br>• processes involved in the main types of mass movement: slides, flows and creeps.<br>The study of at least **two** mass movement events to illustrate:<br>• the interaction of physical and human factors in causing the hazard events;<br>• the resulting impacts (environmental, social and economic);<br>• the human reaction in both short term (emergency rescue) and long term (planning and management). |

| *Questions for Investigation* | *Key Ideas* | *Content* |
|---|---|---|
| What are the hazards associated with flooding? | Flood risk reflects a combination of physical and human factors and these vary from place to place. Flooding has a range of environmental and social impacts on the areas affected, which create a range of human responses to the hazard. | The study of **one** river and **one** coastal area prone to flooding to illustrate:<br>• the physical factors involved (including height, relief, drainage regime, climate, vegetation, rock type);<br>• the human factors involved (including settlement building, farming, deforestation, drainage);<br>• the resulting impacts (environmental, social and economic) of flooding;<br>• the human reaction in both the short term (emergency rescue) and long term (planning and management). |
| What are the hazards associated with earthquake and volcanic activity? | Earthquakes and volcanic eruptions are caused by plate tectonics and bring distinctive impacts to an area and these vary from place to place.<br>Earthquakes and volcanic eruptions have a range of environmental and social impacts on the areas affected, which create a range of human responses to the hazard. | The study of an earthquake and of a volcanic eruption to illustrate:<br>• the tectonic processes involved in creating these hazards;<br>• scale and types of impacts (environmental, social and economic), together with the concept of primary (initial impacts – destruction, casualties, landslides, fires) and secondary impacts (including disease, infrastructure problems, resettlement);<br>• the human reaction in both the short term (emergency rescue) and long term (planning & management). |
| Why do the impacts on human activity of such hazards vary over time and location? | The degree of impact on an area reflects its level of economic and technological development as well as the population density. Impacts can vary over time from immediate to long term. | The study of contrasting examples to illustrate:<br>• a contrast between countries at either end of the development continuum and between rural and urban areas, to compare the impacts of, and reactions to, at least **two** contrasting types of earth hazards;<br>• a comparison of impacts over short and long time periods for at least **two** contrasting types of earth hazards. |

| *Questions for Investigation* | *Key Ideas* | *Content* |
| --- | --- | --- |
| How can hazards be managed to reduce their impacts? | There are various ways to manage or reduce the impacts of hazards. | The study of different approaches to managing earth hazards to illustrate:<br>• the extent to which earth hazards are predictable;<br>• the management strategies used to reduce the possible impact of a hazard;<br>• the effectiveness of managing earth hazards. |

Key Concepts:

- The nature of hazards varies with location
- The nature of hazards changes over time and space
- Earth hazards consist of a variety of interdependent and interconnected activities and processes
- Physical geography and human activity are interdependent and their interaction can produce hazards
- The impact of such hazards varies over time and given location
- Populations and environments respond in a variety of ways to hazards
- The management of hazards results in opportunities and challenges

Associated Skills:

- Research into hazard events
- Analysis of a variety of types of image
- Map work at a variety of scales, eg hazard mapping
- Statistical analysis, eg analysing patterns and severity of hazard
- Use and application of GIS and other modern technology, eg forecasting of earthquakes and eruptions

**Option A2: Ecosystems and environments under threat**

*Candidates need to have a knowledge and understanding of the meaning of 'ecosystems' and 'environment'; that a variety of them can be defined; their variability; and that their sustainability is under threat from short- and long-term impacts of natural and human factors. Case studies should be selected from different areas and at scales appropriate to the question being examined.*

| Ecosystems and environments under threat | | |
|---|---|---|
| *Questions for Investigation* | *Key Ideas* | *Content* |
| What are the main components of ecosystems and environments and how do they change over time? | Ecosystems and environments are systems in which a number of components (physical and human) interact. Environments are subject to constant change as the physical conditions and human activities operating upon them change. | The study of ecosystems to illustrate:<br>• the concept of open and closed systems;<br>• the interconnections between stores and flows in an ecosystem, including energy flows;<br>• how change occurs in an ecosystem as a result of the interaction of physical and human factors. |
| What factors give the chosen ecosystem or environment its unique characteristics? | It is the interaction of the physical and human factors that create distinctive environments and lead to change within them. | The study of at least **one** local ecosystem or environment, eg woodland, dunes or a marsh, to illustrate:<br>• the main stores and flows within the ecosystems;<br>• the main physical factors influencing the chosen environment (including: microclimate, soil, relief, drainage) and how it may develop with time;<br>• the main human influences on the chosen ecosystem or environment (including: conservation, pollution, agriculture, settlement) and how these may generate change with time. |

| *Questions for Investigation* | *Key Ideas* | *Content* |
|---|---|---|
| In what ways are physical environments under threat from human activity? | Human activity poses threats to physical environments in both planned and unintended ways. | The study of at least **one** local ecosystem or environment, eg woodland, dunes or a marsh, to illustrate:<br>• the threats to, and the impacts on, the physical environment posed by a range of human activities (including: agriculture/ forestry, settlement, transport, industry and mineral extraction);<br>• the role that conservation can play in reducing the threats to the environment. |
| Why does the impact of human activity on the physical environment vary over time and location? | The impact of human activity on environments varies as areas develop, and varies between different areas of the world at different stages of economic and technological development. | The study of the contrast between countries at either end of the development continuum to illustrate:<br>• the different ways human activity can impact on physical environments (both positive and negative);<br>• why the impact on physical environments may be increasing or decreasing with economic, social and technological development. |
| How can physical environments be managed to ensure sustainability? | When human activity impacts on physical environments they may need to be managed in order to be sustainable. | The study of at least **one** example of sustainable environmental management of a located physical environment to illustrate:<br>• the ways in which physical environments can/may be managed (including conservation, planning controls, restricted use). |

Key Concepts:

- The nature of threatened ecosystems and the threat posed to them varies with location and over time
- Ecosystems and environments of this nature consist of a variety of interdependent and interconnected activities and processes
- Ecosystems have distinctive characteristics resulting from the interaction of a variety of factors
- Human activity and environmental factors often threaten ecosystems for a variety of reasons
- The impact of human activity varies over time and location
- There are diverse ways of managing an ecosystem in a sustainable way

Associated Skills:

- Research and/or fieldwork to investigate ecosystem characteristics and processes and how these are threatened
- Analysis of a variety of types of image, eg analysis of changes in land use patterns in threatened ecosystems
- Map work at a variety of scales, eg soil maps
- Statistical analysis, including charts and graphs
- Use and application of GIS and other modern technology, eg in measuring, recording and analysing flows in the ecosystem or environment

### Option A3: Climatic hazards

*Candidates need to have a knowledge and understanding of the meaning of ‘hazards’, their degree of predictability and that they make both short- and long-term impacts on an environment and community. Case studies should be selected from different areas and at scales appropriate to the hazards being examined.*

| Climatic hazards | | |
|---|---|---|
| *Questions for Investigation* | *Key Ideas* | *Content* |
| What conditions lead to tropical storms (hurricanes, typhoons or cyclones) and tornadoes and in what ways do they represent a hazard to people? | Tropical storms and tornadoes form and develop under particular atmospheric conditions to become major hazards. These hazards have serious environmental, social and economic impacts upon the areas they affect. | The study of the development of tropical storms and tornadoes to illustrate:<br>• the atmospheric and surface conditions that give rise to their development;<br>• an understanding, with examples, of how such systems develop;<br>• through examples, the hazards they present to particular areas and the impacts that these hazards can have. |
| How do atmospheric systems cause heavy snowfall, intense cold spells, heatwaves and drought and in what ways do they represent a hazard to people? | Atmospheric systems (anticyclones and depressions) can produce extreme weather under certain circumstances, resulting in hazards for people. These hazards have serious environmental, social and economic impacts upon the areas they affect. | The study of high and low pressure systems and air masses to illustrate:<br>• the formation of these hazards, ie heavy snowfall, frost, drought;<br>• how they represent hazards to people through blizzards, cold spells, heatwaves and droughts;<br>• the impacts associated with these weather features for named areas at the local, regional and global scale, including impacts on:<br>– transport;<br>– agriculture and forestry;<br>– health;<br>– economic activity. |

| *Questions for Investigation* | *Key Ideas* | *Content* |
|---|---|---|
| Why do the impacts of climatic hazards vary over time and location? | The degree of impact on an area is related to a variety of factors, including the level of economic and technological development and population density. Impacts can vary over time from immediate to long term. | The study of contrasting examples to illustrate:<br>• a contrast between countries at either end of the development continuum, rural and urban areas, coastal and inland areas, to compare both the impacts of and reactions to at least **two** contrasting types of climatic hazards;<br>• how impacts can vary over short and long time periods for at least **two** contrasting types of climatic hazards. |
| What can humans do to reduce the impact of climatic hazards? | There are a variety of ways to manage or reduce the impacts of climatic hazards. | The study of different approaches to managing atmospheric hazards to illustrate:<br>• the extent to which climatic hazards can be predicted;<br>• different management strategies to reduce their impacts. |
| In what ways do human activities create climatic hazards? | Human activities may impact on the global climate to create particular climatic hazards. | The study of the causes and effects of global warming and global dimming.<br><br>The study for **one** named area of the causes of, impacts on and solutions to, *either* acid rain *or* photochemical smog. |

Key Concepts:

- The nature and impact of hazards vary with space, location and over time
- Climatic hazards consist of a variety of interdependent and interconnected activities and processes
- Physical geography and human activity are interdependent and can together produce hazards
- The impact of such hazards varies over time and location
- Populations and environments respond in a variety of ways to hazards
- The management of hazards results in opportunities and challenges

Associated Skills:

- Research into hazard events and their impact
- Analysis of a variety of types of image
- Map work at a variety of scales, eg climatic hazard mapping
- Statistical analysis including charts and graphs, eg plotting changes in weather observations over time
- Use and application of GIS and other modern technology, eg tracking and forecasting hurricanes, analysis of effect of global warming over time

## Section B: Economic Issues

At least **one** topic must be chosen from this section.

**Option B1: Population and resources**

*Candidates need to have a knowledge and understanding of the meaning of 'resources', their variability in supply, type and locational extent. Population, in turn, should be seen both as a stimulation for the development and exploitation of resources and as consumers of resources. Case studies should be selected from different areas and at scales appropriate to the resources being examined.*

| Population and resources | | |
|---|---|---|
| *Questions for Investigation* | *Key Ideas* | *Content* |
| How and why does the number and rate of growth of population vary over time and space? | Population is dynamic and changes in response to a number of demographic, social, economic and political factors. The factors vary from place to place. | The study of how populations grow over time to illustrate:<br>• the roles of natural increase and net migration;<br>• how population growth is related to concepts of overpopulation and underpopulation;<br>• global contrasts in population growth;<br>• how the rate of growth is changing over time. |
| How can resources be defined and classified? | There are a variety of ways of defining and classifying resources including by source, by use and extent of renewability. | The study of different types of resource to illustrate:<br>• the differences between renewable, non-renewable, flow and semi-renewable resources;<br>• how changes in technology and society may result in changes in the definition of resources. |

| | | |
|---|---|---|
| What factors affect the supply and use of resources? | The supply and use of resources is determined by a combination of physical and socio-economic factors. | The study of at least **two** resources including **one** non-energy resource, eg mineral, foodstuff, fish, forestry, scenery, to illustrate:<br>• the physical factors (including climate, geology, water, soil, vegetation) influencing their supply and use;<br>• the human factors (including technology, capital, transport, population, industry, energy/power supplies, agriculture) influencing their supply and use;<br>• how and why these factors have changed with time. |

| *Questions for Investigation* | *Key Ideas* | *Content* |
|---|---|---|
| Why does the demand for resources vary with time and location? | Different parts of the world have differing demands (in terms of quantities and types of resources) and these change with time and development. | The study of the contrast between countries at either end of the development continuum in their resource supply and use to illustrate:<br>• the link between population size/growth and standard of living and the demand for different resources over time;<br>• the different patterns of demand in MEDC, NIC and LEDC and how these change with population growth and the rate of development. |
| In what ways does human activity attempt to manage the demand and supply of resources and development? | Both the demand for, and supply of, resources need to be planned and managed to achieve a sustainable system. | The study of contrasting types of management and planning strategies used to balance demand and supply for at least **two** different resources, to include:<br>• at least **one** case study of attempts to make resource development sustainable. |

Key Concepts:

- The nature of populations and their demand for resources vary with location
- The nature of populations and demand for resources and their interrelationship change over time and space
- The use and development of resources consist of a variety of interdependent and interconnected activities and processes
- Population changes in number and characteristics over time
- Various dynamic environmental factors influence the demand for and supply of resources
- Demand for and supply of resources vary over time and with location
- The management of resources results in opportunities and challenges
- There are diverse strategies for managing resources in a sustainable way

Associated Skills:

- Research and/or fieldwork into population characteristics and/or resources
- Analysis of a variety of types of image
- Map work at a variety of scales
- Statistical analysis including charts and graphs, eg population pyramids, and analysis of dependency ratios
- Use and application of GIS and other modern technology, eg remote sensing of mineral resources

**Option B2: Globalisation**

*Candidates need to have a knowledge and understanding of the meaning of 'globalisation', why it has developed and its implications for communities and the environment. Case studies should be selected from different areas and at scales appropriate to the issues being examined.*

| Globalisation | | |
|---|---|---|
| *Questions for Investigation* | *Key Ideas* | *Content* |
| What is meant by the term 'globalisation' and why is it occurring? | There are marked advantages for economic activity in working at a global rather than local scale. | The study of the processes of globalisation to illustrate:<br>• the meanings of globalisation in both economic and cultural terms;<br>• the range of factors responsible for this process and the possible future trends. |
| What are the issues associated with globalisation? | Globalisation of economic activity may bring advantages and disadvantages to various areas. These impacts may be environmental, economic, social or political. | The study of the impact of globalisation to illustrate:<br>• the environmental, economic, social and political benefits and problems created in both a NIC and a MEDC;<br>• whether globalisation is increasing or narrowing the 'development gap' (with the aid of statistical analysis). |
| What are transnational corporations (TNCs) and what is their contribution to the countries in which they operate? | Transnational corporations may create both positive and negative impacts on an area. | The study of TNCs to illustrate:<br>• how TNCs may be defined and the ways in which they have developed over time;<br>• **two** case studies of contrasting spatial and organisational structures;<br>• the advantages and disadvantages to countries at either end of the development continuum of at least **one** TNC operation. |

| | | |
|---|---|---|
| How far do international trade and aid influence global patterns of production? | Trade and aid both support and hinder the broader balance of the world's patterns of production. | The study of global trade patterns to illustrate:<br>• the structure, direction and impact of trade for an example of each of a LEDC, NIC and MEDC;<br>• the role of international trade negotiations and agreements.<br>The study of global patterns of aid to illustrate:<br>• the different types of aid;<br>• the advantages and disadvantages of aid for both donor and recipient countries;<br>• examples of short-term emergency aid and examples of long-term development aid. |
| How can governments evaluate and manage the impact of globalisation? | Governments vary in their responses to the impacts of globalisation and are increasingly looking to reduce the harmful impacts. | The study of the different ways of measuring and evaluating the impact of globalisation through a case study of how at least **one** country is managing the impacts of globalisation on its economy and society. |

Key Concepts:

- The nature of globalisation and interrelationships change over time and space
- Globalisation consists of a variety of interdependent and interconnected activities and processes
- The nature and extent of globalisation varies with location
- There are distinctive and diverse features of globalisation and they can also change over time
- Various opportunities and challenges are produced by globalisation
- Transnational corporations, trade and aid all influence global patterns of interdependence and interconnection
- Governments try to manage the diverse nature of challenges and impact of globalisation

Associated Skills:

- Research and/or fieldwork into globalisation features and/or TNCs
- Analysis of a variety of types of image
- Map work at a variety of scales
- Statistical analysis including charts and graphs, eg use of indexes of development
- Use and application of GIS and other modern technology, eg role of Sat Nav and global positioning technologies

**Option B3: Development and inequalities**

*Candidates need to have a knowledge and understanding of the meaning of 'inequalities'; why they may develop; and their implications for communities and the environment. Case studies should be selected from different areas and at scales appropriate to the issues being examined. Gender inequalities should be considered throughout, where appropriate.*

| Development and inequalities | | |
|---|---|---|
| *Questions for Investigation* | *Key Ideas* | *Content* |
| In what ways do countries vary in their levels of economic development and quality of life? | Countries vary in their levels of economic development and this, in turn, influences the quality of life (such as standard of living) of their citizens. | The study of global patterns of economic development and quality of life to illustrate:<br>• different ways of measuring the level of development and quality of life (both quantitative and qualitative);<br>• the contrast in the level of development and the quality of life between LEDCs, NICs and MEDCs (with the aid of statistical analysis and case studies). |
| Why do levels of economic development vary and how can they lead to inequalities? | Various factors influence the rate and level of development and this in turn may increase or decrease economic and social inequalities. | The study of the relative level of development of countries to illustrate:<br>• the factors (physical, economic, social, political and historical) that influence the relative level of economic development of a country;<br>• how economic development can increase or decrease various inequalities between countries and within **one** named country. |
| To what extent is the 'Development Gap' increasing or decreasing? | Some areas are finding it very difficult to develop economically so the inequality gap between the richer and poorer areas is increasing; whilst others are developing rapidly, narrowing the gap. | The study of the concept of the 'Development Gap' to illustrate:<br>• what the 'Development Gap' is and why it exists (models such as the Core-periphery, Friedmann and Rostow are helpful);<br>• the factors (physical, economic, social and political) that may be increasing or decreasing this 'Gap'. |

| *Questions for Investigation* | *Key Ideas* | *Content* |
| --- | --- | --- |
| In what ways do economic inequalities influence social and environmental issues? | Economic inequalities may result in social and environmental conditions also becoming unequal. | The study of variations in social and environmental conditions to illustrate:<br>• variations in pollution (air, water, solid, noise, etc) in both a MEDC and a NIC;<br>• the economic and social inequalities within **one** named region or large city resulting from the interlinking of economic and social factors. |
| To what extent can social and economic inequalities be reduced? | It is important to reduce extreme inequalities and various approaches have been tried with differing degrees of success. | The study of the management of social and economic inequalities to illustrate:<br>• the variety of methods that can be employed to tackle social and economic inequalities and their impacts (including the use of law, education, planning, subsidies, taxation);<br>• the reasons for, and the methods used in, reducing social and economic inequalities in **one** named country. |

Key Concepts:

- The development process consists of a variety of interdependent and interconnected activities and processes
- The development process has diverse characteristics and they change over time
- The nature of development and the resulting inequalities vary with location
- The nature of development and inequalities and their interrelationships change over time and space
- Development impacts on the environment and quality of life and can lead to inequalities
- Inequalities are dynamic and, in turn, influence social and environmental issues
- A variety of attempts are being made to manage or reduce inequalities

Associated Skills:

- Research and/or fieldwork into the scale and type of inequalities
- Analysis of a variety of types of image
- Map work at a variety of scales
- Statistical analysis including charts and graphs, eg quality of life index
- Use and application of GIS and other modern technology, eg use and processing of census data to identify patterns of deprivation

## 3.4 A2 Unit F764: *Geographical Skills*

The aims of this unit are for candidates to develop:

- knowledge and an understanding of the process of geographical research, including fieldwork;
- the skills necessary to complete a piece of individual geographical research;
- the use of technology, eg GIS, remote sensing, etc as research tools;
- a knowledge and understanding of the potential of ICT and its relevance to geographical change;
- the ability to select and use appropriate GIS skills and techniques to explore geographical issues including decision-making and problem-solving;
- an awareness of the problems involved in undertaking individual geographical investigation/research;
- an understanding that interpretation and evaluation of research results should reflect the links and connections between diverse elements of geography.

This Unit is designed to be synoptic. Candidates will use skills in geographical research and investigations/fieldwork acquired during AS and A2. Candidates need to have undertaken individual research on a geographical topic of their choice, following the stages in investigation listed in the table below. This individual research/investigation should be based on any of the topics addressed in Units F761, F762 and F763. It should provide clear evidence of extension and synthesis of understanding and skills.

| *Stages in Investigation* | *Key Ideas* | *Content* |
|---|---|---|
| **1** Identify a suitable geographical question or hypothesis for investigation. | A successful geographical investigation is dependent upon identifying a clear geographical question at a scale that is practical in research terms. | The question/hypotheses should be:<br>• at a suitable scale;<br>• capable of research;<br>• clearly defined and of a clear geographical nature;<br>• based upon wider geographical theories, ideas, concepts or processes. |
| **2** Develop a plan and strategy for conducting the investigation. | A successful geographical investigation requires careful planning, which balances the needs for accuracy and reliability against limitations imposed by time, resources and the environment in which the investigation is being conducted. | This stage requires:<br>• the identification of the data needed to examine the question/hypothesis posed;<br>• the establishment of appropriate strategies and methods for collecting the necessary data (including sampling where appropriate);<br>• an understanding of the limitations imposed by resources and time;<br>• an appreciation of potential risks in undertaking the research, and methods of minimising the risk. |
| **3** Collect and record data appropriate to the geographical question or hypothesis. | A successful geographical investigation is based upon thorough methods of data collection and recording, which consider accuracy and reliability in relation to the data being collected | This stage requires:<br>• the use of primary[1] and secondary data[2] as appropriate to the question/hypothesis posed;<br>• a description and explanation of different ways of collecting/recording data;<br>• an awareness of the need for accuracy and reliability before, during and after the process of data collection. |

[1] Primary data are defined as unprocessed information, which means information collected through fieldwork

| *Stages in Investigation* | *Key Ideas* | *Content* |
|---|---|---|
| **4** Present the data collected in appropriate forms. | A successful geographical investigation involves the selection of techniques that are appropriate to the data collected, and their presentation to a high standard. | This stage requires:<br>• the use of appropriate techniques to present the data collected (including maps, diagrams, annotated photos and graphs);<br>• the logical organisation of the presented material in relation to the analysis;<br>• presentation of the material to a high standard relevant to the question/hypothesis posed. |
| **5** Analyse and interpret the data. | A successful geographical investigation involves a variety of analytical approaches, ranging from the purely descriptive to detailed analysis, involving attempts to explain patterns that have been identified. | This stage requires:<br>• descriptions of the findings shown by the data presentation;<br>• where appropriate, analysis of the data further using statistical techniques[3];<br>• interpretation of the results in relation to the original question/hypothesis posed;<br>• explanations of patterns found and of any anomalous results. |
| **6** Present a summary of the findings and an evaluation of the investigation | A successful geographical investigation involves a clear summary of the findings and an evaluation of the success of the investigation in relation to wider geographical ideas and the methods employed and data collected. | This stage requires:<br>• the use of evidence presented in previous sections to provide a clear conclusion, which relates back specifically to the original question/hypothesis posed;<br>• an evaluation of the extent to which the study supports or otherwise the general geographical theories, ideas or concepts being studied;<br>• an evaluation of the limitations of the study in terms of the methods used and the data collected and suggestions for possible improvements. |

## 4.1 AS GCE Scheme of Assessment

*Geographical skills* are assessed within both units. These include the ability to describe and interpret features, trends and patterns from a variety of geographical sources including:

- OS maps and thematic maps;
- maps presenting statistical data;
- data tables;
- photographs, aerial photographs and satellite and other images (including GIS);
- graphs of various types (pie, bar, line, scatter, histograms);
- diagrams (flow charts, spider diagrams, sketch diagrams and maps).

| AS GCE Geography (H083) |
| --- |
| AS Unit F761: *Managing Physical Environments* |